

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE DEPARTMENTEN & INFORMATION
TECHNOLOGY

SPRING 2025

Information Security
Assignment 03

Semester: VI

Due Date: 19th June-2025

Batch: 2022F

Max Marks: 04

Instructions:

- Submit a **hand written** assignment.

Q#1: Solve the simultaneous congruence's with Chinese remainder Theorem.

$$x \equiv 6 \pmod{11},$$

$$x \equiv 13 \pmod{16},$$

$$x \equiv 9 \pmod{21},$$

$$x \equiv 19 \pmod{25}.$$

Q#3: Find the totient of 12,13, 14, 15. Using Euler totient function.

Q#4: Generate 5 random numbers using Linear congruential number generator where $a=7$, $c=0$, $m=32$, and seed $=1$. Also show the period.

Q#5: find the first 5 bit sequence (random numbers) using Blum Blum Shub by choosing any two prime numbers other than we solved in class (can not use $p=7$, $q=11$). Choose a random number 'S' which is relatively prime to 'n'.

Q#6: Assume $k=[4567]$ plain text $P=[8542]$ whereas state vector $S=[01234567]$. Using RC4 algorithm show key generation, encryption and decryption process.

GOOD LUCK